A Shared LSTM Forecast for 3-D UAV Planning with Coupled MPC–DWA in Dynamic Obstacle Fields

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***Abstract*—First, a UAV must avoid moving obstacles in three dimensions even when turns, acceleration and pauses make their motion difficult to predict. Second, the method couples model predictive control (MPC) and the dynamic window approach (DWA) through one causal long short-term memory (LSTM) forecast. MPC uses it to optimize a route; DWA screens reachable actions and bounds MPC's first command. Then, in a frozen 500-scene comparison with a 1-s DWA screen, shared LSTM achieved collision-free goal arrival in 936/1,500 runs (62.4%), versus 836/1,500 (55.7%) when only DWA received LSTM. The paired difference was 6.7 percentage points (95% scene-cluster confidence interval, 4.2–9.2). In a separate 100-scene exploratory evaluation with a 3-s DWA screen, sharing reached 73.7%; its contrast with DWA-only LSTM was inconclusive. Standalone MPC-CV reached 92.0%, the highest rate in that evaluation. Finally, the simulations show a gain from forecast sharing under the frozen 1-s DWA protocol, while standalone MPC-CV led the separate 3-s evaluation.**

***Index Terms*—UAV, 3D path planning, dynamic obstacles, LSTM, model predictive control, dynamic window approach**

I. Introduction

Unmanned aerial vehicles (UAVs) must avoid obstacles that move during flight. Turns, acceleration and pauses make their recent positions unreliable guides to what happens next. Model predictive control (MPC) optimizes a short future route; the dynamic window approach (DWA) chooses a feasible action for the next step [1], [2]. Recent UAV planners also use predicted obstacle motion in MPC [3], [4]. We test whether one causal long short-term memory (LSTM) forecast, shared by MPC and DWA, improves the coupled planner relative to one-layer forecast input, constant-velocity (CV) prediction and standalone controllers. We also compare with a project-local latest-position-hold reconstruction of the MPC–DWA baseline, because the source paper does not specify an executable dynamic-obstacle predictor [5]. The shared forecast connects route optimization to DWA's reachable-action screening and local safety gates.

II. Problem Formulation

We model the UAV as a collision sphere in a bounded three-dimensional workspace with a start, goal and moving spherical obstacles. Each obstacle has an observed center, radius and time-varying path; future ground-truth positions are hidden from the planner. At each update, it chooses a dynamically feasible action to reach the goal without intersecting an obstacle along the simulated motion. Figure 1 shows the scene geometry; recorded planner paths are compared separately.

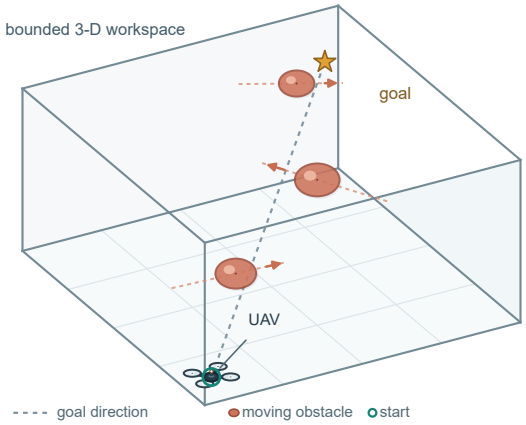


Figure 1. Problem geometry: the quadrotor is centered at the start and moves toward the goal among spherical obstacles following prescribed paths (arrows). The dashed line marks the direct start-to-goal direction, not a simulated route; planner trajectories appear separately in Figure 4.

III. Planning Method

With the scene defined, the planner uses one causal obstacle forecast to align route optimization with near-term action screening.

A. Core equations

The following compact relations define the state update, obstacle forecast and quantities used by the planners. The UAV is represented by its position, speed, pitch and yaw. The signed distance between its collision sphere and obstacle j is

$$\begin{aligned} \mathbf{h}_t &= [\cos\theta_t \cos\psi_t, \cos\theta_t \sin\psi_t, \sin\theta_t]^{\mathsf{T}}, \\ \mathbf{p}_{t+1} &= \mathbf{p}_t + \Delta t\, v_t \mathbf{h}_t, \\ v_{t+1} &= v_t + \frac{\Delta t}{\tau_v}(v_{c,t} - v_t), \\ \theta_{t+1} &= \theta_t + \Delta t\, \omega_{\theta,t}, \quad \psi_{t+1} = \psi_t + \Delta t\, \omega_{\psi,t}, \\ \hat{d}_{j,k} &= \|\mathbf{p}_{t+k|t} - \hat{\mathbf{o}}_{j,t+k|t}\|_2 - r_u - r_j. \end{aligned} \tag{1}$$

Here t is the current step, k a future step, and $j = 1, \ldots, M$ indexes the M obstacles. The unit direction $\mathbf{h}_t$ follows the

UAV's pitch θ_t and yaw ψ_t. The update period is $\Delta t = 0.1$ s; τ_v is the speed-response time; $v_c, \omega_\theta, \omega_\psi$ are commanded speed, pitch rate and yaw rate. In the clearance term, $\mathbf{p}_{t+k|t}$ and $\hat{\mathbf{o}}_{j,t+k|t}$ are the predicted UAV position and obstacle centre at step k. The UAV and obstacle radii are r_u and r_j. Positive $\hat{d}_{j,k}$ means separation, zero contact, and negative overlap.

At each update, the causal LSTM [6] maps ten obstacle observations to thirty 3-D offsets from the latest observed centre:

$$\begin{aligned} \mathbf{q}_{j,s} &= [\mathbf{o}_{j,s}^{\mathsf{T}}, \mathbf{v}_{j,s}^{\mathsf{T}}, \cos\chi_{j,s}, \sin\chi_{j,s}]^{\mathsf{T}}, \\ \hat{\mathbf{o}}_{j,t+k|t}^{\mathrm{LSTM}} &= \mathbf{o}_{j,t}^{\mathrm{obs}} + [f_\Theta(\mathbf{q}_{j,t-9:t})]_k. \end{aligned} \tag{2}$$

Here s indexes history and $k = 1, \ldots, 30$ indexes the forecast steps. The position $\mathbf{o}$ and velocity $\mathbf{v}$ in $\mathbf{q}$ are observed obstacle inputs, followed by sine and cosine of heading χ; thus $\mathbf{q}$ has eight features. The trained network f_Θ uses learned weights Θ and outputs a 3-D offset for each future step; $[\cdot]_k$ denotes offset k. Ten observations span 0.9 s and the 30 outputs span 3 s. The forecast is causal and each offset is anchored to the latest observed centre, not cumulatively summed. At startup, the earliest state fills missing history. CV baselines extrapolate from the latest two observed positions (zero initial velocity when only one is available); position hold repeats the latest centre. Training details are in Supplementary Methods.

MPC uses the forecast over its 30-step horizon to minimize goal error, control effort and changes, route deviation, and obstacle-clearance penalties:

$$\begin{aligned} \min_{\mathbf{u}_{1:30}} J_{\mathrm{MPC}} = &\sum_{k=0}^{29} w_p \|\mathbf{p}_{t+k|t} - \mathbf{g}\|_2^2 \\ &+ w_f \|\mathbf{p}_{t+30|t} - \mathbf{g}\|_2^2 + w_r J_r \\ &+ \sum_{k=1}^{30} w_u \|\mathbf{u}_k\|_2^2 + w_{\Delta u} \sum_{k=2}^{30} \|\mathbf{u}_k - \mathbf{u}_{k-1}\|_2^2 \\ &+ \sum_{k=1}^{30} \sum_j \left(w_o [-\hat{d}_{j,k}]_+^2 + w_s [d_{\mathrm{soft}} - \hat{d}_{j,k}]_+^2 \right), \\ \text{s.t. } &\mathbf{x}_{t+k|t} = f(\mathbf{x}_{t+k-1|t}, \mathbf{u}_k), \quad \mathbf{u}_1 \in \mathcal{U}_t \text{ if } \mathcal{A}_t \neq \varnothing. \end{aligned} \tag{3}$$

Here $\mathbf{u}_{1:30}$ denotes the sequence of 30 future commands; $\mathbf{x} = [\mathbf{p}^{\mathsf{T}}, \theta, \psi, v]^{\mathsf{T}}$ is the position–pitch–yaw–speed state, $\mathbf{u}_k = [v_{c,k}, \omega_{\theta,k}, \omega_{\psi,k}]^{\mathsf{T}}$ is command k, $\mathbf{g}$ is the goal, and f is the discrete state transition in Eq. (1). J_r sums squared cross-track distances from the direct start-to-goal line. The weights $w_p, w_f, w_r, w_u, w_{\Delta u}, w_o, w_s$ apply to stage goal error, terminal error, route deviation, control effort, command changes, overlap and clearance below $d_{\mathrm{soft}} = 0.35$ m. The positive-part function is $[x]_+ = \max(0, x)$. Physical command and rate limits remain active; because the implementation squares native command units, the effort term is a heuristic cost rather than physical energy. Full geometric definitions, weights and limits are in Supplementary Methods S1.1. These clearance penalties are soft, so they do not certify safety by themselves.

Trust allocates DWA samples between guided and exploratory actions using safety, goal progress, agreement with the previous MPC guide, and control smoothness:

$$\begin{aligned} \tau &= \sigma_s \tau_{\mathrm{sa}} + \sigma_g \tau_g + \sigma_t \tau_{\mathrm{ref}} + \sigma_m \tau_{\mathrm{sm}}, \\ B &= \mathrm{round}[300(1 + \alpha e^{-\beta\tau})], \\ B_g &= \mathrm{round}(B\tau), \quad B_e = B - B_g. \end{aligned} \tag{4}$$

The normalized scores $\tau_{\mathrm{sa}}, \tau_g, \tau_{\mathrm{ref}}, \tau_{\mathrm{sm}}$ denote safety, goal progress, reference agreement and smoothness; their nonnegative weights $\sigma_s, \sigma_g, \sigma_t, \sigma_m$ sum to one. B is the sample budget, divided into MPC-guided B_g and exploratory B_e actions. Higher trust concentrates more of a smaller budget near the MPC guide; lower trust increases exploration. The dimensionless $\alpha = 1$ and $\beta = 3$ are fixed implementation settings, not selected in the 4,320-episode Stage-1 sweep, which selected the weights only. Six candidate weight vectors were compared for the 3-s evaluation, selecting $[\sigma_s, \sigma_g, \sigma_t, \sigma_m] = [0.15, 0.15, 0.15, 0.55]$ (equal-arm macro-average 0.8042); this same-data selection is descriptive, not held-out evidence. The frozen 1-s benchmark retained its inherited weights $[0.35, 0.30, 0.15, 0.20]$. Full score definitions and parameters are in Supplementary Methods S1.2.

DWA rejects unsafe rollouts, scores the survivors, and uses their control range to bound MPC's first input:

$$\begin{aligned} \mathcal{A}_t = \{&b \in \mathcal{B}_t : d_{\min}^{(b)} > 0.25 \text{ m}, \ \mathcal{C}_b = 1, \\ &(v_{\mathrm{end}}^{(b)})^2 \leq 2 a_{\max} d_{\min}^{(b)}\}, \quad b^* = \arg\max_{b \in \mathcal{A}_t} S_b, \\ u_{i,t}^- &= \max\left(u_i^{\min}, \min_{b \in \mathcal{A}_t} u_i^{(b)} - \delta \right), \\ u_{i,t}^+ &= \min\left(u_i^{\max}, \max_{b \in \mathcal{A}_t} u_i^{(b)} + \delta \right), \\ \mathcal{U}_t &= \prod_{i=1}^{3} [u_{i,t}^-, u_{i,t}^+], \quad \delta = 0.1. \end{aligned} \tag{5}$$

Here $\mathcal{B}_t$ is the sampled set, and $d_{\min}^{(b)}$ is candidate b's minimum clearance over the active DWA screen and all obstacles. The frozen benchmark screens the first 10 forecast states (1 s); the separate exploratory evaluation screens all 30 (3 s). MPC uses a 30-step (3-s) horizon in both. The indicator $\mathcal{C}_b = 1$ means the rollout remains inside the flight corridor. The terminal speed is $v_{\mathrm{end}}^{(b)}$ and available deceleration is $a_{\max} = 1.0$ m s^{-2}. S_b ranks MPC-endpoint agreement, heading, speed, obstacle clearance and vertical profile. For a nonempty safe set $\mathcal{A}_t$, its per-coordinate extrema, expanded by $\delta = 0.1$ in each control coordinate and clipped to its physical limits $u_i^{\min}, u_i^{\max}$, form $\mathcal{U}_t$; this numerical margin is an implementation heuristic because the controls have different units. The DWA action is executed unless optional lateral recentering is triggered; the adjusted rollout is re-screened, and failure reverts to the original screened action. The envelope constrains only MPC's first command and is not a safety certificate. If the normal safe set is empty, emergency actions are screened; if none passes, the simulator stops. Full scores and fallback logic are in Supplementary Methods S1.2–S1.4.

B. How the shared forecast is used

Every 0.1 s, the causal LSTM produces a time-aligned 3-s forecast for both layers. MPC uses all 30 future positions in its route cost; DWA screens the first 10 positions in the frozen benchmark and all 30 in the separate exploratory evaluation. DWA's passing controls bound MPC's first command on re-solve, while the screened DWA action is executed. The same predicted encounters therefore guide route optimization and

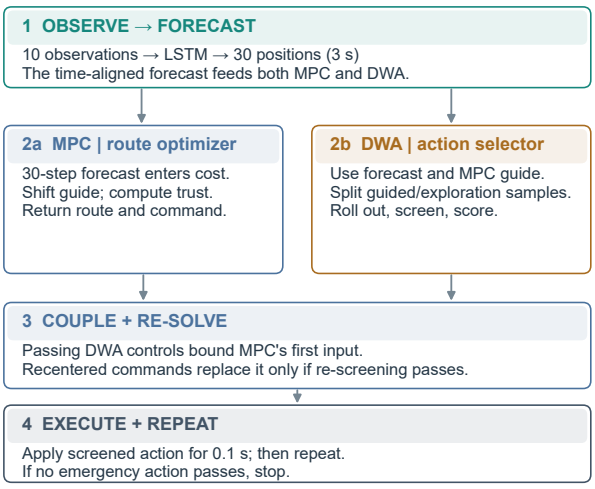


Figure 2. Stepwise modules in one shared-forecast update. The same causal LSTM forecast feeds the MPC route optimizer and DWA action selector. Passing DWA controls bound MPC's first input; a triggered recentering is used only after re-screening. The screened DWA or emergency action is executed, with fail-closed stop if no emergency action passes.

reachable-action screening. In the historical DWA-only arm, MPC holds the latest obstacle positions; in the 3-s one-layer ablation, MPC uses its configured CV input. Figure 2 summarizes the cycle; Supplementary Methods S1.4 gives the full fallback logic.

IV. Simulation and Results

We report the frozen matched comparison first and the separate exploratory evaluation second; their protocols and scene banks are not pooled.

Under the frozen 1-s DWA/3-s MPC protocol with inherited weights, 500 scenes and three matched seeds compared shared LSTM with DWA-only LSTM. DWA received LSTM forecasts in both arms; only the shared arm also supplied them to MPC. Safe arrival was 936/1,500 (62.4%) versus 836/1,500 (55.7%), a paired gain of 6.7 percentage points (95% scene-cluster CI, 4.2–9.2). These 3,000 runs are part of the 7,500-run benchmark, not an extra sample. In the same protocol, standalone DWA+LSTM and MPC+LSTM reached 82.4% and 95.2%, respectively, while the latest-position-hold reconstruction reached 2.2% (Fig. 3a; Supplementary Methods S4). Thus, forecast sharing improved on DWA-only input but did not outperform the independent controllers overall.

The separate exploratory evaluation comprised 2,100 episodes: seven arms, 100 distinct scenes and three matched seeds per scene. It used 0.1-s updates, 3-s MPC/DWA horizons and a 650-step limit. The arms were standalone DWA-CV and MPC-CV, coupled MPC–DWA-CV, LSTM to MPC only, LSTM to DWA only, shared LSTM and latest-position hold. Arrival required goal tolerance without swept-path contact; seeds were averaged within scene, and 20,000 scene-cluster bootstrap resamples formed simultaneous max-—t— intervals for six shared-versus-comparator contrasts. Shared LSTM reached in 221/300 runs (73.7%), compared with 276/300 (92.0%) for standalone MPC-CV, the highest rate; MPC-CV also had 15 swept-contact runs. Its difference from DWA-only LSTM was inconclusive; it exceeded position hold by 16.3 points (95% CI, 5.1–27.6) but was 18.3 points below MPC-CV (95% CI, -29.6 to -7.0). This evaluation also does not establish an overall best controller (Fig. 3b; Supplementary Table S12).

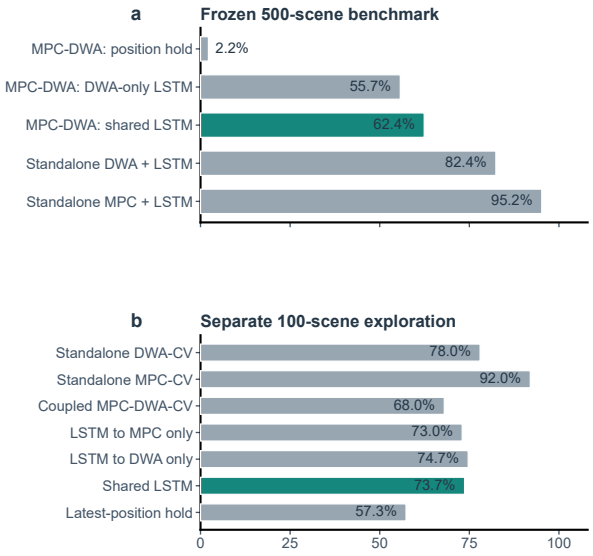


Figure 3. Collision-free goal-arrival rates in two separate protocols. (a) Frozen five-controller benchmark: 500 scenes, three matched seeds, $n = 1{,}500$ per controller and a 1-s DWA screen. (b) Exploratory seven-controller evaluation: 100 distinct scenes, three matched seeds, $n = 300$ per controller and a 3-s screen. Bars show point estimates; paired scene-cluster intervals are reported in the text and supplement. Swept-contact counts follow the displayed method order: 624, 2, 0, 0, 36 in (a), and 0, 15, 0, 0, 0, 50 in (b). The highest-rate method differs by protocol; panels are not pooled.

A matched route from the separate seven-method evaluation (scene 72, sampling seed 2) is shown in Fig. 4. Shared LSTM and DWA-CV arrived safely in this example; the other five did not achieve collision-free goal arrival. This single case illustrates route geometry, not the aggregate ranking.

V. Discussion

The two protocols indicate a protocol-specific benefit: forecast sharing raised safe arrival in the frozen 1-s screen comparison, but not in the 3-s exploratory comparison; standalone MPC-CV had the highest rate there. Because scene banks, trust weights and horizons also differed, these studies do not isolate the source of that difference. LSTM uses simulator-recorded obstacle velocity and heading, whereas CV estimates velocity from positions, so the contrast concerns configured input pipelines as well as predictors. Prescribed tracks, point forecasts and no flight tests limit generalization.

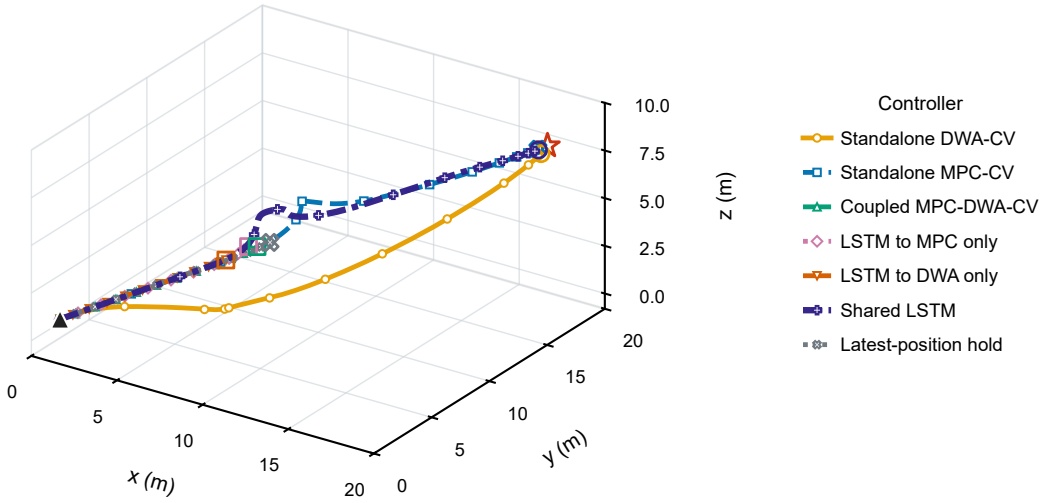


Figure 4. All seven recorded trajectories from scene 72, seed 2, overlaid in one 3-D scene. Color, line style and sparse recorded-state markers identify each controller. Shared LSTM and DWA-CV safely arrived; MPC-CV reached goal tolerance but had swept contact; the other five stopped or contacted earlier. Paths end at logged states without smoothing, interpolation or extension. Triangle/star mark start/goal; terminal circle/square/X denote safe arrival, no-contact non-arrival or swept contact (X is not the contact location). This single example is not aggregate evidence; rates are in Fig. 3b.

VI. Conclusion

In simulation, shared LSTM increased collision-free arrival by 6.7 percentage points over DWA-only input in the frozen 500-scene, 1-s DWA comparison. In the separate 3-s evaluation, standalone MPC-CV had the highest rate, and the shared-versus-DWA-only contrast was inconclusive.

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